Name: _____

Class: _____Date:_____

LESSON 7. ORDER OF OPERATIONS

Calculate the value of each expression (Exercises 1.62 to 1.65):

1.62. a) $6 + 9 + 8 - 11$; b) $3 \cdot 4 \cdot 5 \cdot 6 : 8$.

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1.63. a) $4 \cdot 10^3 + 7 \cdot 10^2 + 2 \cdot 10$;

b) $48 - 3 \cdot 1^{200} + 2 \cdot 5 \cdot 5^2$;

c) $3 \cdot 5^3 + 4 \cdot 6 - 64 \cdot 5 + 9$.

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1.64. a) $[(52 - 4) : 12]^{2+3}$;

b) $3^3 + 2 \cdot \{10 + 3 \cdot [2 \cdot (7 - 3) + 1] + 2\} + 1$.

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1.65. a) $P = 3x^3 + 2x^2 + 4x + 5$ when $x = 2$;

b) $P = a^2 - 2ab + b^2$ when $a = 5$; $b = 3$.

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1.66. Find the natural number x such that:

a) $12x + 36 = 8 \cdot 3^2 + 4 \cdot (2 + 3 + 4)$;

b) $95 - 3x = 3 \cdot 5^2 - 4 \cdot 6 + 96 : 12$.

1.67. At 6:30 a.m., a truck and a motorbike leave A at the same time and travel to B. The truck travels at 48 km/h and the motorbike at 32 km/h. At 8:30 a.m., a car also leaves A for B at 60 km/h.

a) Suppose a second motorbike leaves A for B at the same time as the truck and the first motorbike, travelling at 40 km/h. Write expressions for the distances travelled by the truck, the first motorbike and the second motorbike after t hours. Show that the second motorbike is always exactly halfway between the truck and the first motorbike;

b) Write expressions for the distances travelled by the second motorbike and the car x hours after the car leaves;

c) At what time is the car exactly halfway between the first motorbike and the truck?